

Question	Answer	Mark	Mark scheme
18	Proof	B1	for $\overrightarrow{EH} = -\mathbf{b} + \mathbf{a}$ or $\overrightarrow{HE} = -\mathbf{a} + \mathbf{b}$ or $\overrightarrow{HG} = \mathbf{a} + \mathbf{c}$ or $\overrightarrow{GH} = -\mathbf{c} - \mathbf{a}$
		B1	for $\overrightarrow{BC} = -2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}$ or $\overrightarrow{CB} = 2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}$
		B1	for $\overrightarrow{EF} = \mathbf{b} + \frac{1}{2}(-2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}) (= \mathbf{a} + \mathbf{c})$ or $\overrightarrow{FE} = \frac{1}{2}(2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}) - \mathbf{b} (= -\mathbf{a} - \mathbf{c})$ or $\overrightarrow{FG} = \frac{1}{2}(-2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}) - \mathbf{c} (= -\mathbf{b} + \mathbf{a})$ or $\overrightarrow{GF} = \mathbf{c} + \frac{1}{2}(2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}) (= -\mathbf{a} + \mathbf{b})$
		A1	for completing proof, $\overrightarrow{EH} = \overrightarrow{FG}$ or $\overrightarrow{HG} = \overrightarrow{EF}$ (so $EFGH$ is a parallelogram)